PicoMagnetometer RPi Python code install manual

Pico nT

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Python code for the UKRAA PicoMagnetometer

Set of Python code to run on an RPi4/5 to get, process and present data from the UKRAA PicoMagnetometer

This software was written to suit a specific set-up, feel free to use as you see fit.

Instructions for setting up a Raspberry Pi4/5 are included in the **docs** folder

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Using the code

The code assumes that your UKRAA PicoMagnetometer is the only device using USB that is connected to the RPi4/5 via supplied USB cable and that it is **/dev/ttyACM0** - you can check this by using **ls /dev/tty*** in a terminal window on the RPi4/5 and reviewing the response.

The code assumes username is **pi**.

If **pi** is not the username, then you will need to change all occurrences of '**/home/pi**' to '**/home/*username***' in the python and gnuplot scripts, where ***username*** is the username you have selected for your RPi4/5.

The code assumes one magnetometer connected to the RPi4/5 USB and that it will be connected via **/dev/ttyACM0.**

If there are other devices connected to the RPi and your magnetometer is not **/dev/ttyACM0**, then you will need to change **/dev/ttyACM0** to **/dev/*ttyACMx*** in the **GetDataRawACM0.py** python script, where ***ttyACMx*** is the tty address of you connected magnetometer.

GetDataRawACM0.py is run as a service.

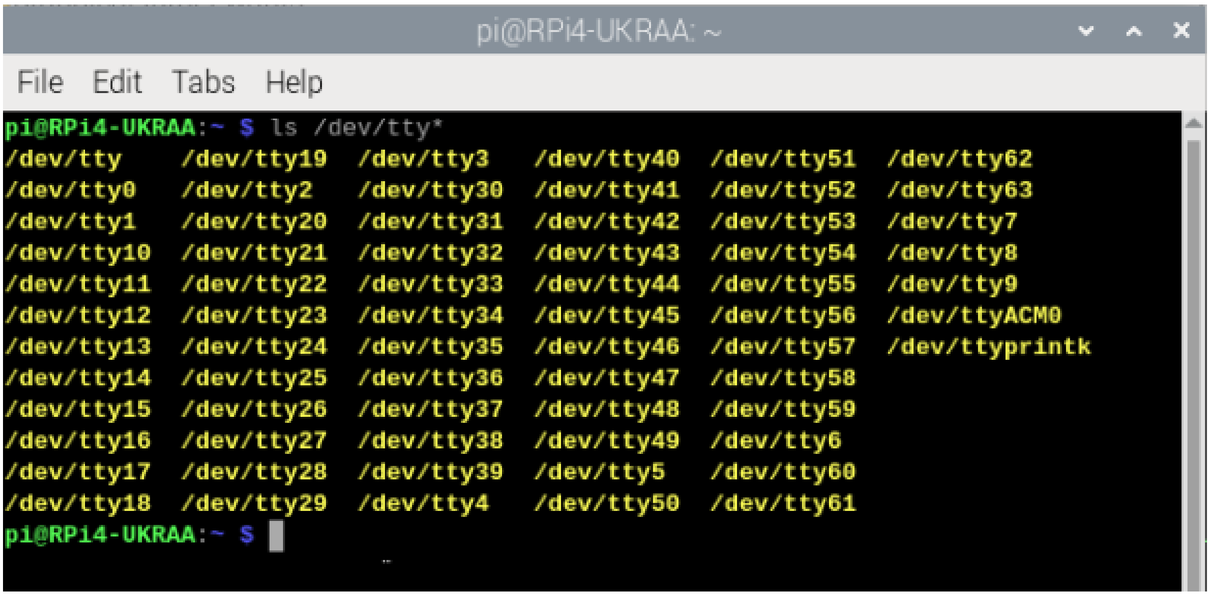
Other scripts (Python and gnuplot) are run from **cron** using shell scripts.

Where is my PicoMagnetometer?

Plug your PicoMagnetometer into any of the RPi USB ports - I normally use one of the blue ports (USB3).

1. Open a terminal window and type the following command and press enter

   ls /dev/tty*



2. You are looking for **/dev/ttyACM0** - this is on the right hand side of the screen shot above.

3. This is the USB address for your attached PicoMagnetometer - if you have more than one magnetometer attached you may see **/dev/ttyACM1** etc.

4. If you do not see **/dev/ttyACM0**, then unplug and plug the PicoMagnetometer back in and try again.

5. As long as you see **/dev/ttyACM0** then you do not have to make any changes to the python scripts, because they are looking for **ACM0**.

Getting the software onto your RPi

1. Log into your Raspberry Pi4/5 using VNC.

2. Open a terminal window, type the following command and press enter

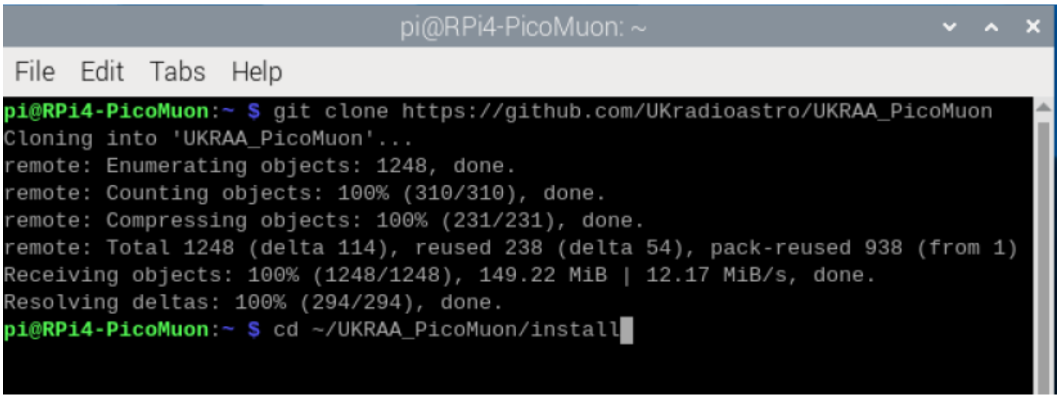
   git clone https://github.com/UKradioastro/UKRAA_Magnetometer

This will download all of the code to the directory **UKRAA_Magnetometer** inside **/home/pi**

Install the software onto your RPi

1. Open a terminal window and type the following command and press **enter**

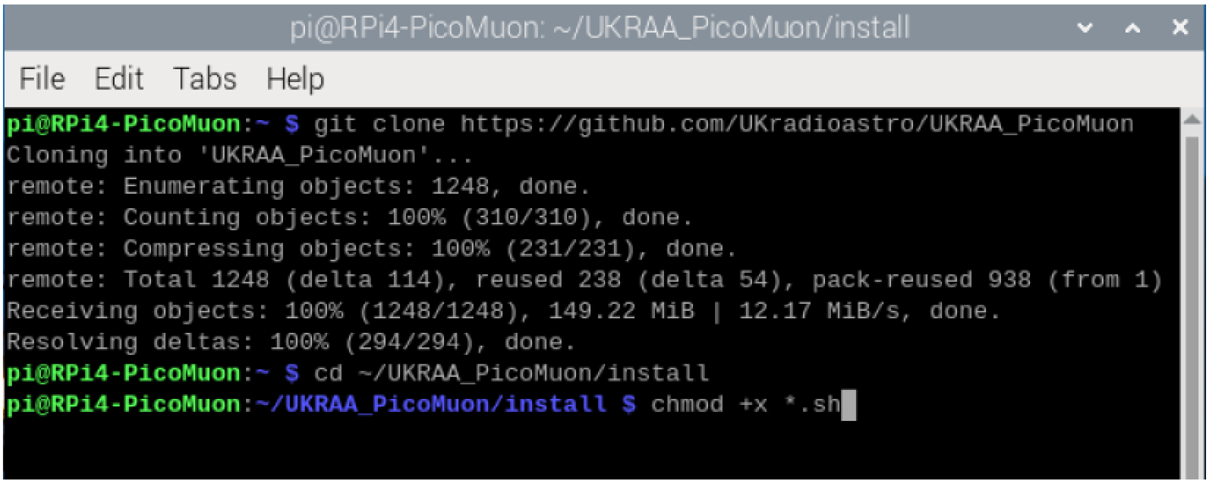
   cd ~/UKRAA_Magnetometer/install



This will take you to the **install** directory inside **/home/pi/UKRAA_Magnetometer**

2. Type the following command and press **enter**

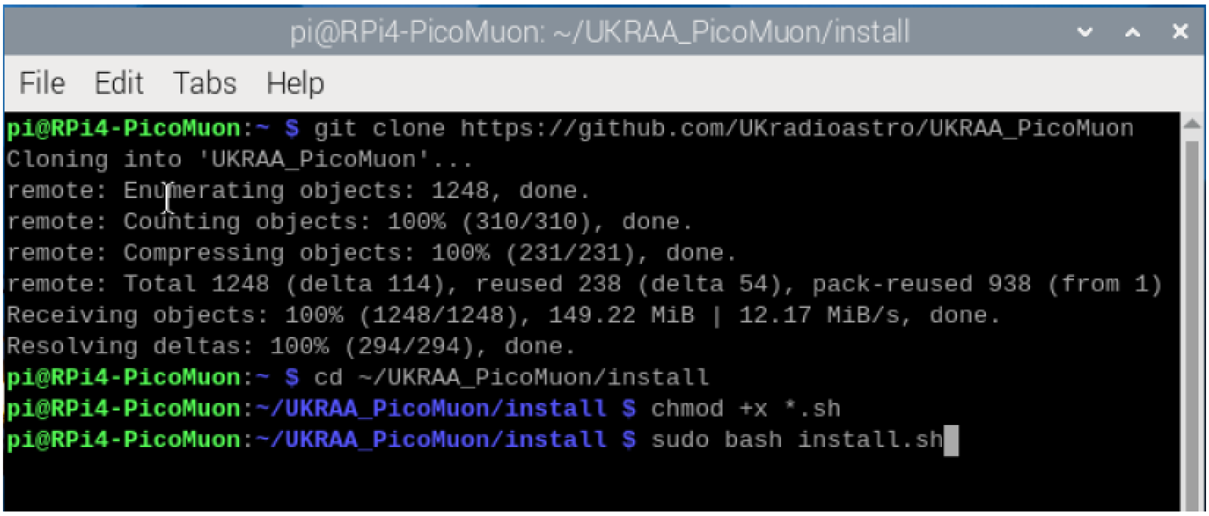
   chmod +x *.sh



This will make the **install.sh** script executable.

3. Type the following command and press **enter**

    sudo bash install.sh



This will run the install script.

There may be occasions during the running of the install script that require you to make a keyboard entry.

When asked **Do you want to continue? [Y/n]** - type **Y** or **y** and press **enter**

That's it!

The code is now set up to run automatically; it will get the data from the PicoMagnetometer, process the data, plot the data and post the plots to your intranet web page.

There are other functions that are customisable by the user, such as **email alerts** and **FTP** to users' external website, that are covered in the **User Manual**.

Optional: run an install-time heartbeat smoke check (disabled by default):

sudo MAGNETOMETER_INSTALL_SMOKE_HEARTBEAT=1 bash install.sh

This runs **--test-heartbeat** once during install and prints a clear **HEARTBEAT_SMOKE_CHECK: PASS** or **HEARTBEAT_SMOKE_CHECK: FAIL** summary.

What does the code do?

The code receives data from the UKRAA PicoMagnetometer via serial over the supplied USB cable and stores the data to the raw data folder.

The raw data will be processed, via CRON, to get magnetic activity per minute, the variation of the X (N->S) and Y (E->W) magnetic field activity per hour and then plot to relevant graphs.

A number of plots will be created:
- X, Y and Z (North/South, East/West and up/down)
- H, D and Z (Local horizontal plane, declination angle and up/down)
- B and I (Total strength of Earth's magnetic field and angle of Earth's magnetic field)

Note: H, D & Z plots together with B & I plots are disabled by default, these plots can be generated by changing the configuration files - see **User Manual: Additional graphs** for more details.

The raw data will also be processed on a continuous 5 minute basis, again via CRON, to produce a rolling 24 hour plot of X, Y and Z magnetic fields and % change of magnetic field for combined X and Y directions. The latter used to predict the potential of visible Aurora activity. Should a threshold level be reached for % change of magnetic field, then the user will receive an email alert of such - see **User Manual: Rolling alert emails** for more details.

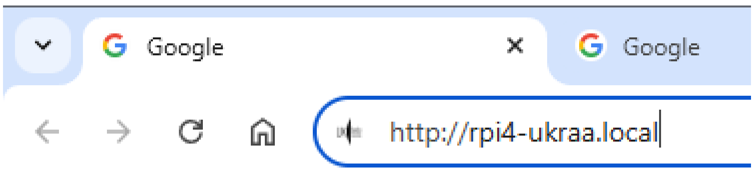
A simple web server and web page is set up on your RPi so that you can view your PicoMagnetometer results on your desktop PC and/or smart phone when connected to your home network.

To access the webpage from your desktop PC or your smart phone…

1. Open your preferred web application (Safari, Chrome, Firefox, etc.).

2. In the search bar type the following and press enter

   http://test-mag.local



This will take you to the web page for your PicoMagnetometer on your local intranet, displaying yesterday's events graphs and rolling 24 hour activity graph.

NOTE: if you have a different **hostname** for your RPi, change the search bar entry to…

http://*hostname*.local

Where ***hostname*** is the hostname for your RPi setup.

Code is also supplied that will enable the user to upload, via FTP, to their host website, should they have one - see **User Manual: FTP service** for more details.

Check PicoMagnetometerACM0 service is running

1. To check the **status** of your service, type the following command and press enter.

   sudo systemctl status PicoMagnetometerACM0.service

2. To **start** your service, type the following command and press enter.

   sudo systemctl start PicoMagnetometerACM0.service

3. To **stop** your service, type the following command and press enter.

   sudo systemctl stop PicoMagnetometerACM0.service

4. To **enable** your service, type the following command and press enter.

   sudo systemctl enable PicoMagnetometerACM0.service

5. To **disable** your service, type the following command and press enter.

   sudo systemctl disable PicoMagnetometerACM0.service

License

Contact us

Please send an e-mail to Magnetometer@ukraa.com

UK Radio Astronomy Association